

Connecting to SDWAN

Please note, this module is optional and is not required to be completed to proceed with this workshop. You may skip to [Network Monitoring](#)

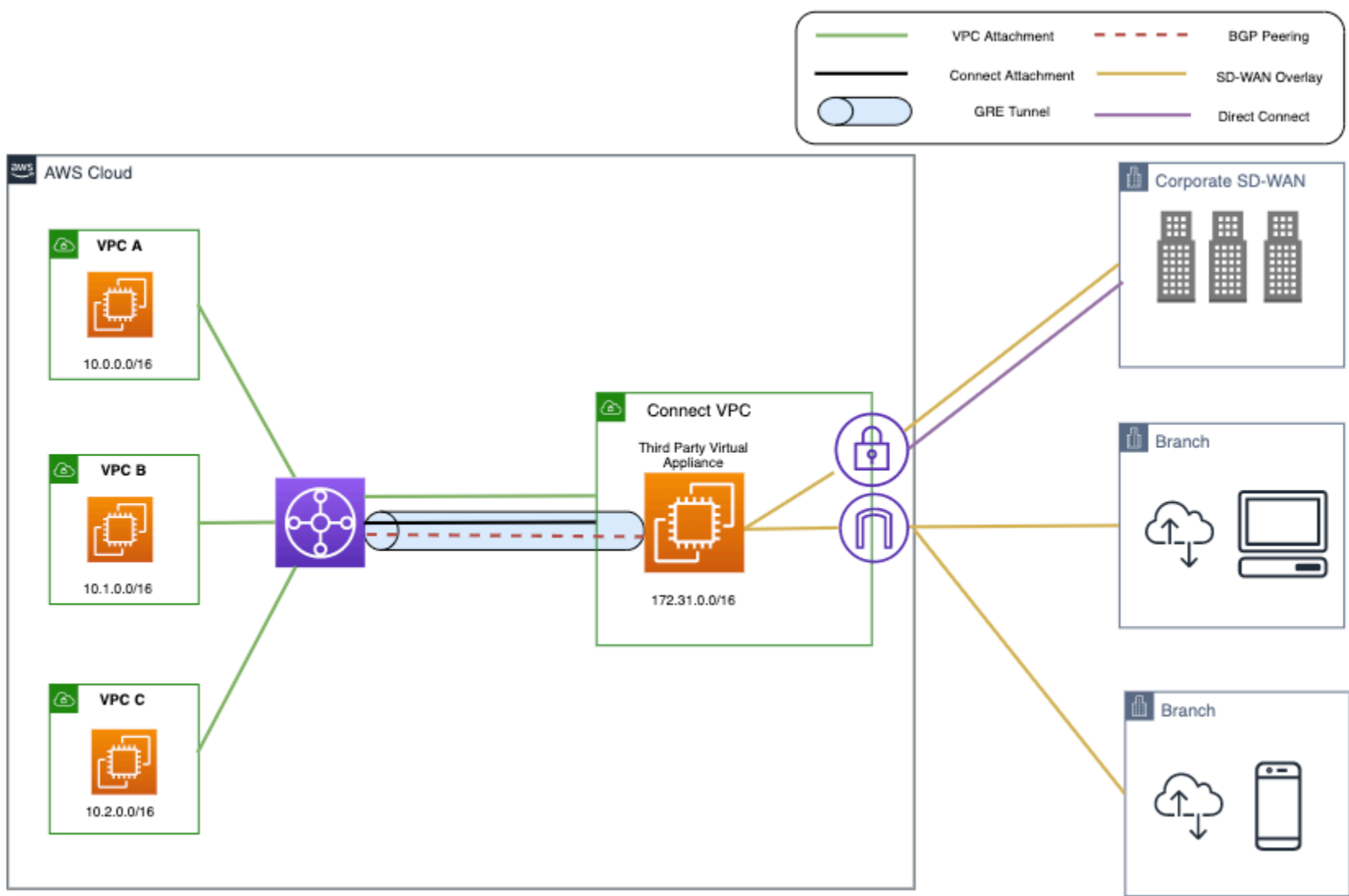
⚠ Elastic IP quota

This module's stack creates two Elastic IP addresses. If the **Connecting to On-Premises** stack is still deployed in this account, first complete only the on-premises-specific steps of the [On-Premises cleanup](#) — delete the Site-to-Site VPN, the On-Premises stack, and the Route 53 Resolver endpoints, stopping before "Continue deleting resources", which removes shared stacks this module still needs. Running both modules at once exceeds the default quota of five Elastic IPs per Region and this stack will fail to deploy.

AWS Transit Gateway Connect simplifies the integration of third-party SD-WAN and networking appliances with AWS by enabling direct BGP peering with Transit Gateway over GRE tunnels. Transit Gateway Connect allows third-party appliances to leverage existing VPC attachments or Direct Connect attachments as a transport layer, allowing higher bandwidth when compared to VPN.

By using GRE tunnels between network appliances and AWS Transit Gateway, organizations can establish dynamic routing with BGP, enabling seamless route exchange without the complexity of manually configuring static routes. This approach enhances scalability and provides flexible connectivity between SDWAN fabric and AWS VPCs over private network.

A Connect peer can only be established over a VPC attachments or Direct Connect attachments — these serve as the underlying transport that carries GRE traffic between your appliance and the Transit Gateway. This enables appliances running in AWS or in your on-premises environment (via Direct Connect) to dynamically exchange routes with the Transit Gateway.



In this lab, we will set up a **third-party network appliance** and integrate it with an existing **AWS Transit Gateway** using a **Transit Gateway Connect Attachment**. This hands-on experience will demonstrate how GRE tunneling and BGP peering enable dynamic route propagation, providing a scalable and efficient solution for integrating SD-WAN appliances with AWS networking services.

To build out our simulated network appliance environment and connect it to AWS, we will:

1. Deploy a VPC containing a third-party network appliance.
2. Create a **Transit Gateway Connect Attachment**.
3. Establish a **GRE tunnel** between the appliance and Transit Gateway.
4. Enable **BGP peering** over the GRE tunnel.
5. Validate connectivity and route propagation.

ⓘ If you are running this lab at an **AWS-led event**, your facilitator has already set the region for you. If you are running it **self-service**, simply use the region you are working in. In either case, the region shown in the screenshots may not match your environment — this will not cause any problems.

If you are running this lab in **your own AWS Account**, it is recommended for all lab resources to be created in **us-east-1** so that the screenshots match your environment. This is not mandatory.

Prerequisites

► If you have not completed the Multiple VPCs sections...

Deploying the Third-Party Network Appliance Environment

Now, we are ready to deploy a provided CloudFormation template to set up the simulated third-party appliance. This instance will act as a network router connecting to AWS via Transit Gateway Connect. The appliance functions as a border router in a BGP-enabled network, enabling it to exchange routing updates with the Transit Gateway over BGP.

1. Load the following CloudFormation template:

[CloudFormation Template](#)

2. Enter the **Stack name** ThirdPartyAppliance-TGWConnect.
3. Remaining parameters will use the default values.

Create stack ⓘ

Step 1

● Create stack

Step 2

● Specify stack details

Step 3

○ Configure stack options

Step 4

○ Review and create

Specify stack details

Provide a stack name

Stack name

ThirdPartyAppliance-TGWConnect

Stack name must contain only letters (a-z, A-Z), numbers (0-9), and hyphens (-) and start with a letter. Max 128 characters. Character count: 30/128.

Parameters

Parameters are defined in your template and allow you to input custom values when you create or update a stack.

AMI

AMI ID for the Ubuntu instances

ami-0e1bed4f06a3b463d

InstanceProfile

IAM Instance Profile Name

NetworkingWorkshopInstanceProfile

InstanceType

EC2 instance type for the routers

t3.micro

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4. Click **Next**, then **Create stack**.

Wait for the stack to be created. This process can take up to **5 minutes**.

⚠ Important

If the deployment of the CloudFormation template fails due to the **Elastic IP address** limit being reached and you are using your own AWS account, please refer to the [Elastic IP Quota](#) section in [Prerequisites - Your own AWS Account](#).

Now we are ready to create the Transit Gateway Connect Attachment to integrate the third party appliance with AWS Transit Gateway.

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Accept

Decline

Customize

AWS Networking Workshop

Foundational Topics

Prerequisites

VPC Fundamentals

Multiple VPCs

Security Controls

Connecting to On-Premises

Connecting to SDWAN

Create TGW Connect Attachment

Configure the Third-Party Appliance

Validate Connectivity

Clean Up

Network Monitoring

Network analytics using Amazon Quick Suite

Advanced Topics

Contributors & Revision History

Workshop catalog in AWS Builder Center

Content preferences

Language

English

Create TGW Connect Attachment

In this section, we will create the infrastructure required to establish a Transit Gateway Connect attachment that integrates a third-party router (running in a simulated Appliance VPC) with AWS Transit Gateway over a GRE tunnel. This process enables BGP dynamic routing between the third-party appliance and AWS using Transit Gateway Connect.

We will walk through the following steps:

- Create a **VPC attachment** to connect the Appliance VPC to the Transit Gateway.
- Assign a **CIDR block** to the Transit Gateway (required for Connect peers).
- Create a **Transit Gateway Connect attachment** that uses the VPC attachment as its transport.
- Configure a **Connect peer** to establish the GRE tunnel and BGP session.

Step 1: Create a VPC Attachment to Transit Gateway

In this step, we will create a **VPC attachment** from the Appliance VPC to the existing Transit Gateway. This attachment will be used as a transport layer for the TGW Connect Peer.

- Navigate to the [Transit Gateway Attachments](#) page.
- Click **Create Transit Gateway Attachment**.
- Provide the following parameters:

- Name tag:** Appliance-VPC-Attachment
- Transit Gateway ID:** Select the Transit Gateway **Name**: TGW.
- Attachment type:** VPC.
- VPC ID:** Select the Appliance VPC.
- Subnets:** Select the Appliance-TGW-Subnet (e.g., 10.5.5.0/28).

Availability Zone Requirement for GRE Tunnel

The GRE tunnel between the Transit Gateway and the third-party appliance **must be established in the same Availability Zone** as the subnet used in the **VPC attachment (transport attachment)**.

Transit Gateway delivers GRE packets through the **AZ-local subnet**. If the appliance's ENI is in a different AZ than the attachment subnet, the tunnel will fail to establish.

Refer to the official AWS documentation:
[Transit Gateway Connect – Connect Peer Requirements](#)

- Wait for the status of the attachment to become **Available**.

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Endpoints

Endpoint services

Service networks

Lattice services

Resource

Transit gateway attachments (1/4)

Find transit gateway attachment by attribute or tag

Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID
☐ VPC C Attachment	tgw-attach-03f68b	tgw-0ae0f5	Available	VPC	vpc-0b9dc3
☐ VPC B Attachment	tgw-attach-0633ec	tgw-0ae0f5	Available	VPC	vpc-08ae33
☒ Appliance-VPC-Attachment	tgw-attach-068817	tgw-0ae0f5	Available	VPC	vpc-01ee47
☐ VPC A Attachment	tgw-attach-09f6cd	tgw-0ae0f5	Available	VPC	vpc-071ab8

Transit gateway attachment: tgw-attach-068817cs / Appliance-VPC-Attachment

Details

Flow logs

Tags

Details

Transit gateway attachment ID

tgw-attach-068817cs

Transit gateway ID

tgw-0ae0f5

Transit gateway owner ID

27

Appliance Mode support

Disable

State

Available

Resource owner ID

278

Subnet IDs

subnet-098c75d

Resource type

VPC

Resource ID

vpc-01ee47

Security Group Referencing support

Association state

~

Association route table ID

-

IPv6 support

Disable

Step 2: Add a CIDR to the Transit Gateway

Transit Gateway Connect requires the Transit Gateway to have a CIDR block from which the **Transit Gateway GRE IP** can be assigned.

- Navigate to the [Transit Gateways](#) section in the VPC Console.
- Select your Transit Gateway.
- Select **Actions**, then **Modify transit gateway**.
- Locate the **Transit gateway CIDR blocks** section.
- Enter the following:
 - CIDR block:** 10.10.0.0/24
- Click **Modify transit gateway**.

Important

The transit gateway CIDR blocks must not overlap with the VPC CIDR blocks or with any remote network that is advertised to the transit gateway.

VPC

Transit gateways

tgw-0ae0f5

Modify transit gateway

Details

Transit gateway ID

tgw-0ae0f5

Description - optional

Set the description of your transit gateway to help you identify it in the future.

Immersion Day TGW

Configure the transit gateway

Amazon side Autonomous System Number (ASN)

64512

DNS support

Security Group Referencing support

VPN ECMP support

Default route table association

Default route table propagation

Configure cross-account sharing options

Auto accept shared attachments

Transit gateway CIDR blocks

10.10.0.0/24

10.10.0.0/24

Cancel

Modify transit gateway

Step 3: Create the Transit Gateway Connect Attachment

Next, we will create the **Connect attachment** that enables GRE and BGP connectivity with the third-party appliance.

- Go to the [Transit Gateway Attachments](#) section again.
- Click **Create Transit Gateway Attachment**.
- Provide the following:
 - Name tag:** TGWConnect-Appliance
 - Transit Gateway ID:** Use the same Transit Gateway **Name**: TGW..
 - Attachment type:** Connect
 - Transport Attachment:** Select Appliance-VPC-Attachment .
- Click **Create transit gateway attachment** and wait for the attachment to become **Available** before proceeding.

VPC

Transit gateway attachments

Create transit gateway attachment

Create transit gateway attachment

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details

Name tag - optional

Creates a tag with the key set to Name and the value set to the specified string.

TGWConnect-Appliance

Transit gateway ID

tgw-0ae0f5

Attachment type

Connect

Connect attachment

A connect attachment allows you to establish connection between a transit gateway and the third-party appliances using Generic Routing Encapsulation (GRE) and Border Gateway Protocol (BGP).

Transport attachment ID

Select a transit gateway attachment

tgw-attach-03f68b VPC C Attachment

tgw-attach-0633ec VPC B Attachment

tgw-attach-068817 Appliance-VPC-Attachment

tgw-attach-09f6cd VPC A Attachment

Remove

Cancel

Create transit gateway attachment

Step 4: Add a Connect Peer

Once the Connect attachment is active, we will add a **Connect peer** that represents the GRE tunnel between AWS and the Third-party Appliance.

- On the **Transit gateway attachments** page, click on the **TGWConnect-Appliance** attachment.
- Under **Connect peers**, click **Create connect peer**.
- Provide the following parameters:
 - Name tag:** GRE1-tunnel
 - Transit gateway GRE address:** 10.10.0.1 (must be from the TGW CIDR range)
 - Peer GRE address (GRE outer IP):** 10.5.0.6 (Third Party Appliance ENI1 IP)
 - BGP inside CIDR blocks IPv4:** 169.254.255.0/29

Important

The **inside IP addresses** used for BGP peering must come from a **/29 CIDR block** within the 169.254.0.0/16 range (IPv4). For IPv6, you may optionally use a **/125 CIDR block** from the fd00::/8 range.

The following CIDR blocks are reserved and cannot be used:

- 169.254.0.0/29
- 169.254.1.0/29
- 169.254.2.0/29
- 169.254.3.0/29
- 169.254.4.0/29
- 169.254.5.0/29
- 169.254.169.248/29

- Peer ASN:** 65500 (as configured on Third Party Appliance for you in the prerequisites section)

Important (IBGP Behavior)

If you're using **IBGP** (same ASN on both the Transit Gateway and the peer), **routes advertised by the third-party appliance will not be installed** in the TGW route table unless they were originally learned from an **eBGP** peer and include `next-hop-self`.

- Click **Create**.

VPC

Transit gateway attachments

tgw-attach-0e4792e962160ceed

Create connect peer

Create connect peer

A connect peer is a Generic Routing Encapsulation (GRE) tunnel within which you can establish Border Gateway Protocol (BGP) peering to exchange routes.

Details

Name tag - optional

Creates a tag with the key set to Name and the value set to the specified string.

GRE1-tunnel

Transit gateway ID

tgw-0

Connect attachment ID

tgw-attach-

Configure tunnel options

Customize GRE tunnel addresses and BGP inside CIDR blocks for your connect peer. Unspecified tunnel options will be auto generated.

Transit gateway GRE address - optional

Requires a valid IPv4 or IPv6 address.

10.10.0.1

Peer GRE address

Requires a valid IPv4 or IPv6 address.

10.5.0.6

BGP inside CIDR blocks IPv4

Requires a valid IPv4 CIDR mask.

169.254.255.0/29

BGP inside CIDR blocks IPv6 - optional

Requires a valid IPv6 CIDR mask.

/125 IPv6 CIDR

Peer ASN - optional

Requires a valid BGP ASN.

65500

- Wait for the Connect peer status to become **Available**.

Step 5: Update the Appliance Subnet Route Table to Reach Transit Gateway GRE IP

To allow the Third-Party Appliance to reach the **Transit Gateway GRE IP** (e.g., 10.10.0.1), we must add a route to the **subnet route table** associated with the **network interface** used by the GRE tunnel (Router1ENI1).

In this lab, the GRE-facing ENI of the Third-Party Appliance is in the subnet **Appliance-Public-Subnet** (10.5.0.0/28), which is associated with the **Appliance-RT** route table.

To update the route:

- Navigate to the [Route Tables](#) section in the [VPC Console](#).
- Find the route table named **Appliance-RT**.
- Select the **Routes** tab and click **Edit routes**.
- Click **Add route** and enter:
 - Destination:** 10.10.0.0/24 (or the exact TGW GRE IP as /32, e.g., 10.10.0.1/32)
 - Target:** Select Transit Gateway, then in the 2nd drop box, select the Transit gateway attachment **Appliance-VPC-Attachment**.
- Click **Save changes**.

Important

If this route is not present, the Third-Party Appliance will not be able to send GRE traffic to the Transit Gateway, and the Connect Peer will remain down.

VPC

Route tables

rtb-061a

Edit routes

Edit routes

Destination	Target	Status	Propagated
10.5.0.0/16	local	Active	No
172.20.16.5/32	Peering Connection	Active	No
0.0.0.0/0	Internet Gateway	Active	No
10.10.0.0/24	Transit Gateway	-	No

Add route

Cancel

Preview

Save changes

In the next section, we will start configuring the Third-party Appliance in the simulated SDWAN network with the parameters from the TGW Connect Attachment to create the

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If you agree, AWS and approved third parties will also use cookies to provide useful site features, remember your preferences, and display relevant content, including relevant advertising. To accept or decline all non-essential cookies, choose "Accept" or "Decline." To make more detailed choices, choose "Customize."

Accept

Decline

Customize

Validate Connectivity

In this section, you will validate the dynamic routing configuration using BGP over GRE between AWS Transit Gateway and your simulated Third-Party Appliance.

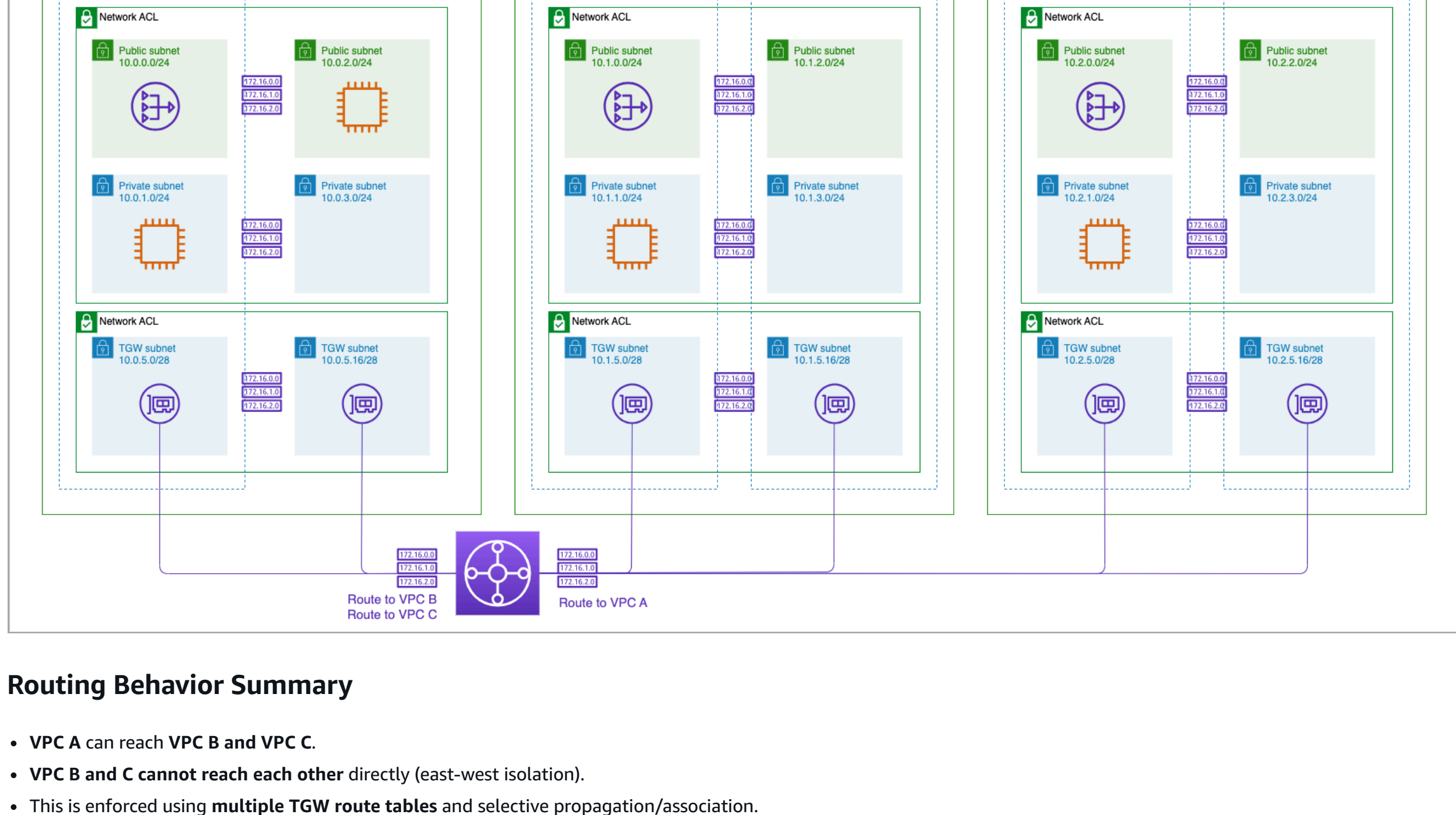
What Has Been Deployed in your account

You have launched a multi-VPC architecture consisting of:

- **Three VPCs** (VPC A, B, and C), each in two Availability Zones.
- Each VPC includes:
 - Public and private subnets
 - TGW subnets for validation
 - EC2 instances for validation
- A centralized **AWS Transit Gateway (TGW)** connecting all VPCs.
- A **shared services design**, where:
 - **VPC A** is the shared VPC.
 - **VPC B** and **VPC C** are isolated from each other.
- A **TGW Connect attachment** and established a GRE tunnel and BGP peering between the TGW and a simulated Third-Party Appliance.

Transit Gateway Routing Domains

The following diagram illustrates the current routing design:



Routing Behavior Summary

- **VPC A** can reach **VPC B** and **VPC C**.
- **VPC B** and **C** cannot reach each other directly (east-west isolation).
- This is enforced using **multiple TGW route tables** and selective propagation/association.
- **BGP** is up between the simulated third-party appliance and TGW using a **TGW Connect attachment**.

Next, we need to configure the TGW route tables to enable full dynamic route exchange.

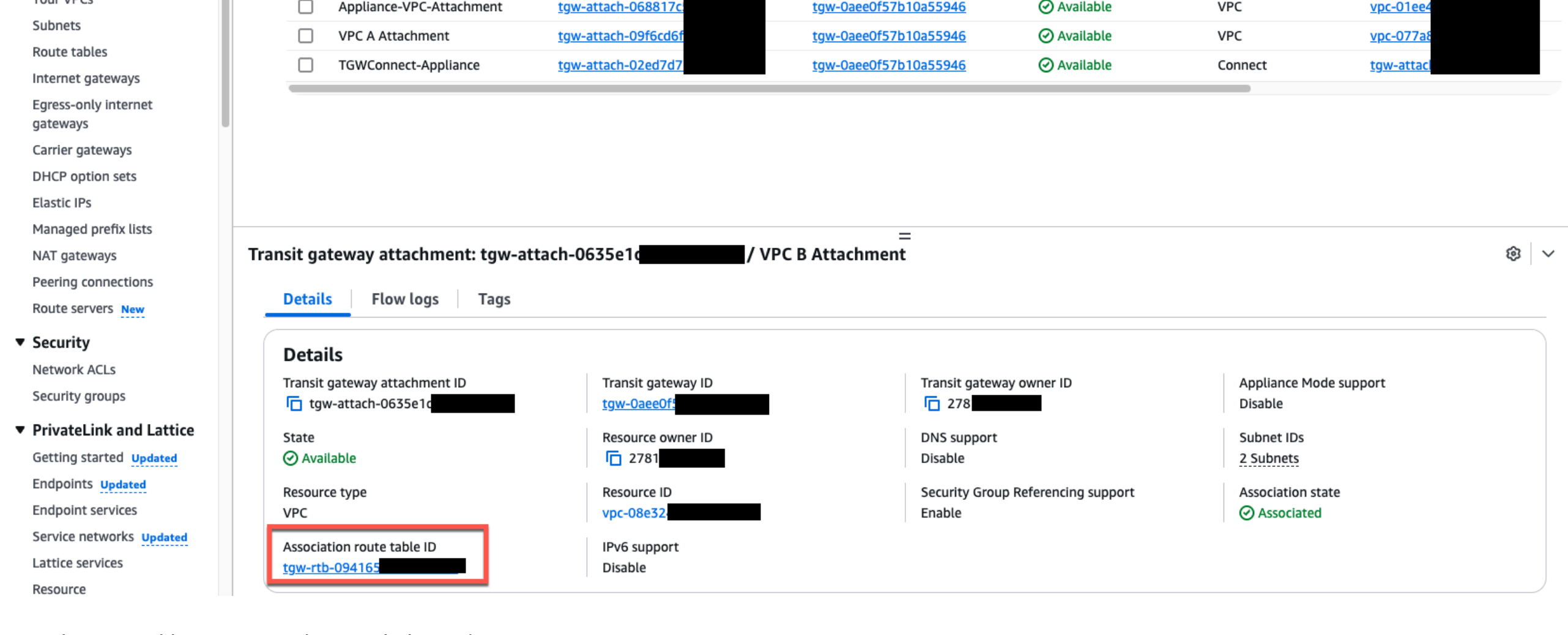
Step 1: Reachability between Branch1 Router and VPC A

To allow the **Third-Party network** to reach resources in **VPC A**, we need to associate the **TGW Connect Attachment** with the same route table used by **VPC B** & **VPC C**. This ensures the appliance can use that route table to locate the next attachment for **VPC A**.

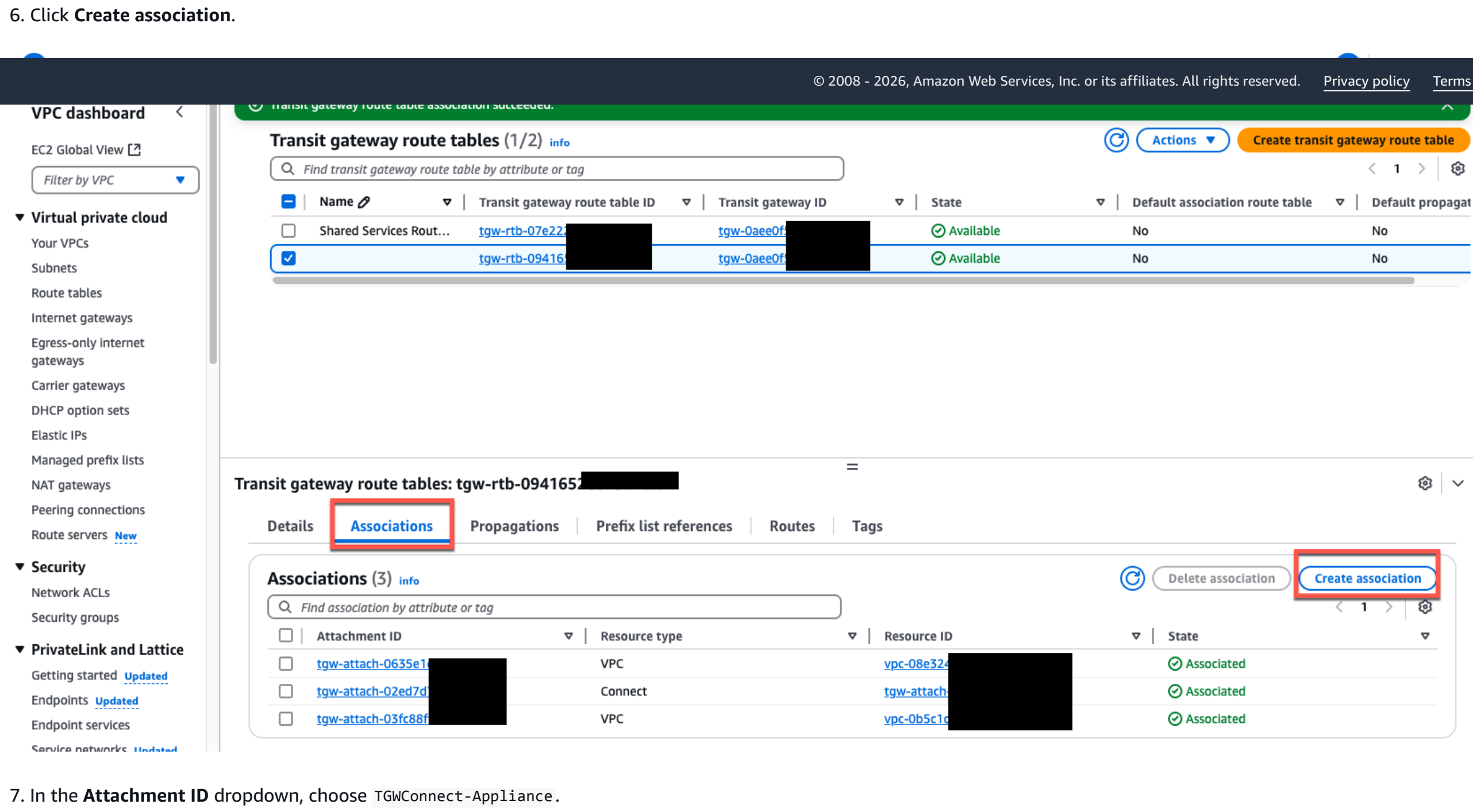
Associate the Connect Attachment

To associate the connect attachment to the same route table associated with VPC B and VPC C, locate and Open the Route Table Used by VPC B Attachment on TGW

1. Open the [Transit Gateway Attachments Console](#).
2. Search for the attachment with **Name tag**: VPC-B-TGW-Attachment.
3. Click to open its details.
4. Scroll to the **Association route table ID**, then click the linked **Route Table ID** to open the associated Transit Gateway route table directly.



5. In the route table page, go to the **Associations** tab.
6. Click **Create association**.



7. In the **Attachment ID** dropdown, choose **TGWConnect-Apppliance**.
8. Click **Create association**.

The **Third-Party Appliance** now uses this route table to send traffic towards **VPC A**. It will not be able to reach **VPC B**, or **VPC C** since it is in the spoke route table that does not have routes to the spoke VPCs.

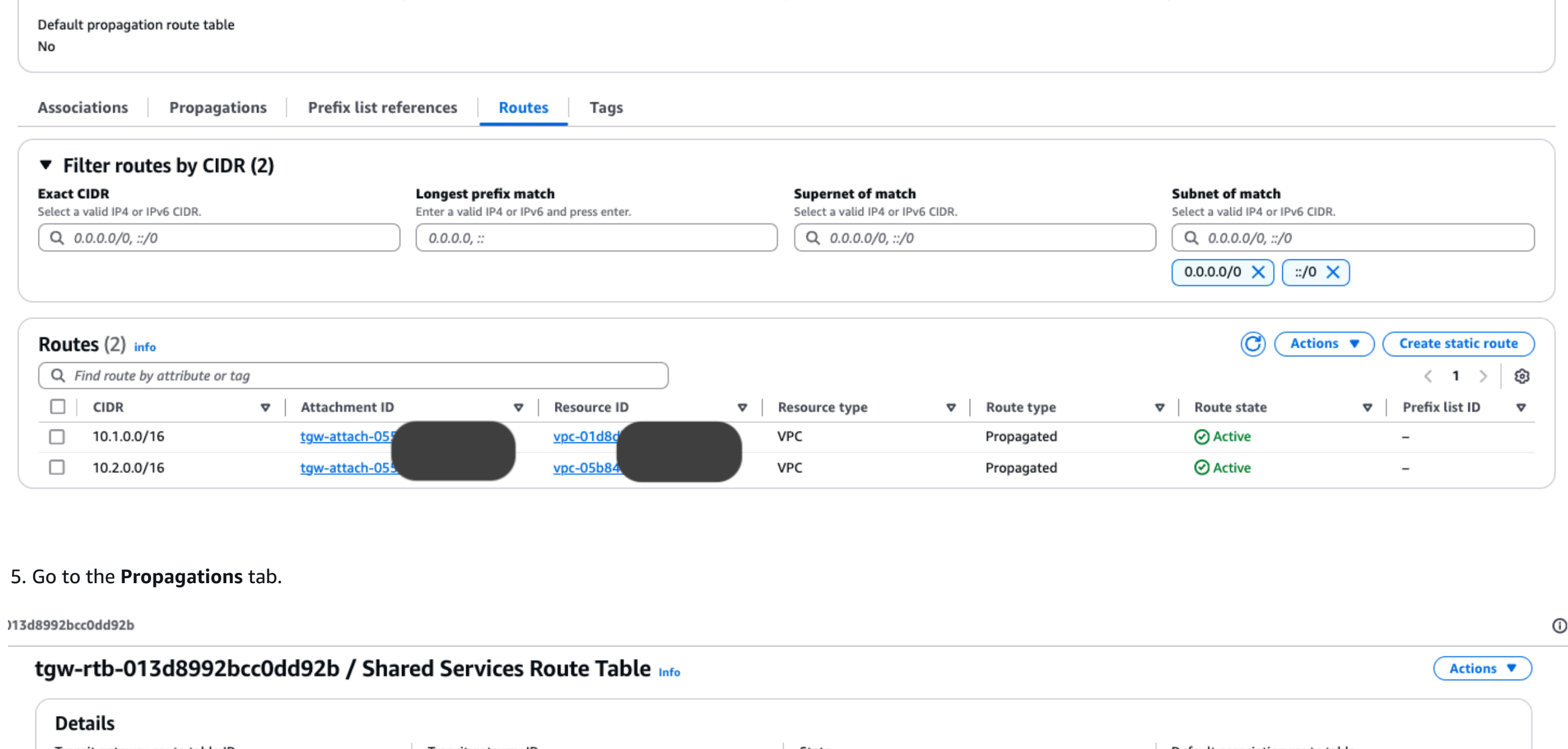
Step 2: Propagate Appliance Routes into the Shared Services Route Table

To allow **VPC A** to send traffic to networks behind the **Third-Party Appliance**, we need to **enable propagation** from the Connect attachment into the **Shared Services Route Table**.

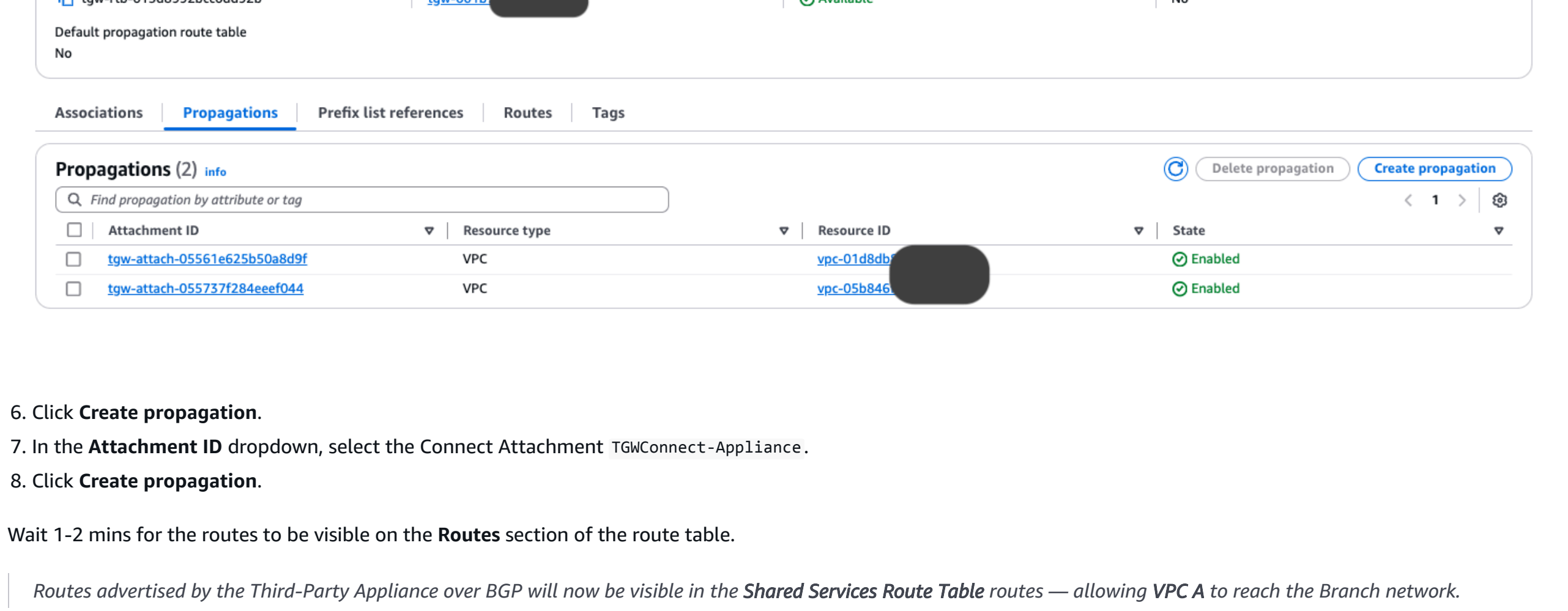
This ensures that **BGP-learned routes** from the appliance are available in the route table used by **VPC A**, enabling return traffic to external destinations.

Enable Propagation in the Shared Services Route Table

1. Open the [Transit Gateway Route Tables Console](#).
2. Locate the route table with **Name tag**: Shared Services Route Table.
3. Click the route table name to open its details.
4. Go to **Routes**. Notice here there are only routes for the **VPC-B:10.1.0.0/16** and **VPC-C:10.2.0.0/16**. We will propagate to this table routes learned from the **Third-Party appliance**.



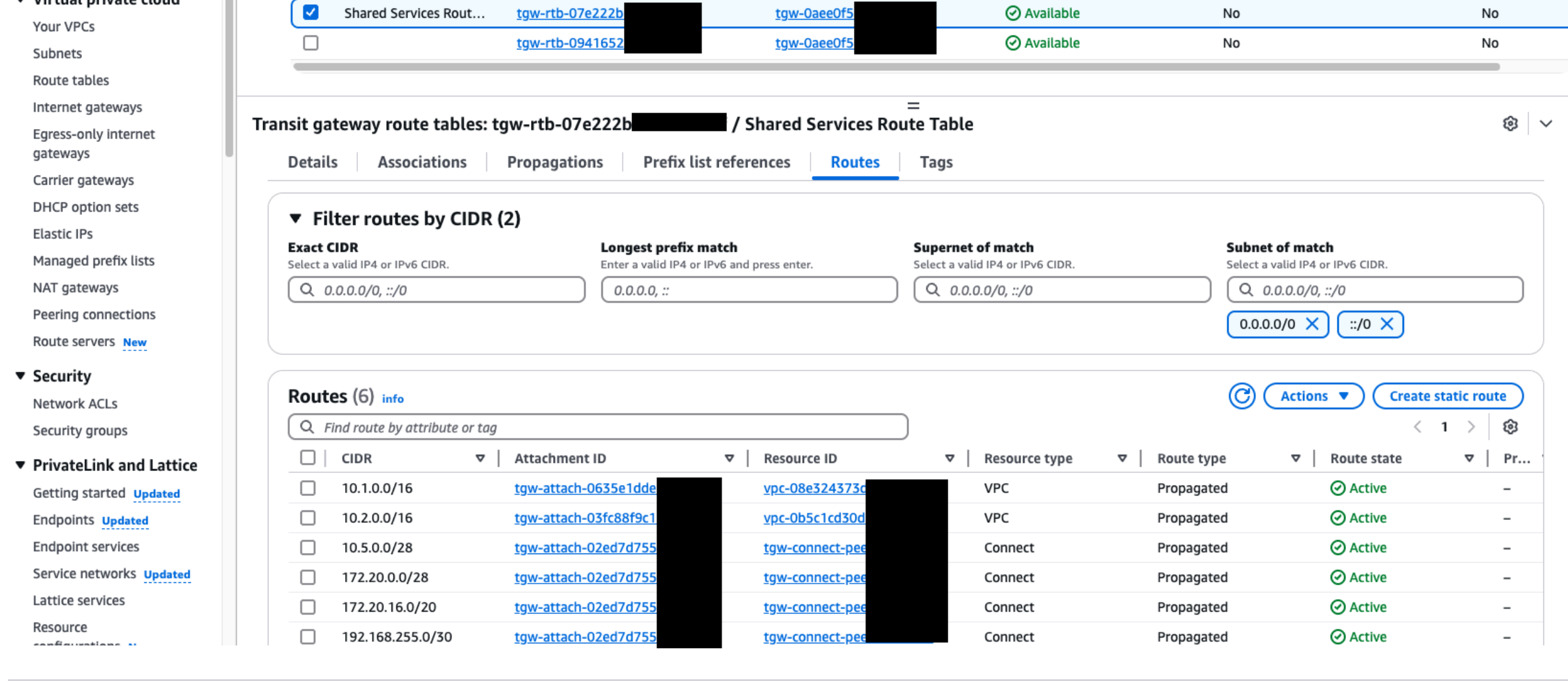
5. Go to the **Propagations** tab.



6. Click **Create propagation**.
7. In the **Attachment ID** dropdown, select the Connect Attachment **TGWConnect-Apppliance**.
8. Click **Create propagation**.

Wait 1-2 mins for the routes to be visible on the **Routes** section of the route table.

Routes advertised by the **Third-Party Appliance** over **BGP** will now be visible in the **Shared Services Route Table** — allowing **VPC A** to reach the **Branch network**.



Testing Strategy

Now that routing is configured, we will validate that routing is working as expected.

Our goal is to ensure:

- The **Third-Party network** can reach resources in **VPC A**.
- The **VPC A** route table learns external routes via the **Connect attachment**.
- The **Branch1 Router** can reach the EC2 instance **VPC A Private AZ1 Server** over **BGP**.

We'll achieve this by associating and propagating the correct route tables, then validating reachability through testing.

!Architecture\\static\\foundational\\images\\BGPOverGRE\\Lab4CnDeployed.png]

Step 3: View BGP-Learned Routes on the Third-Party Appliance

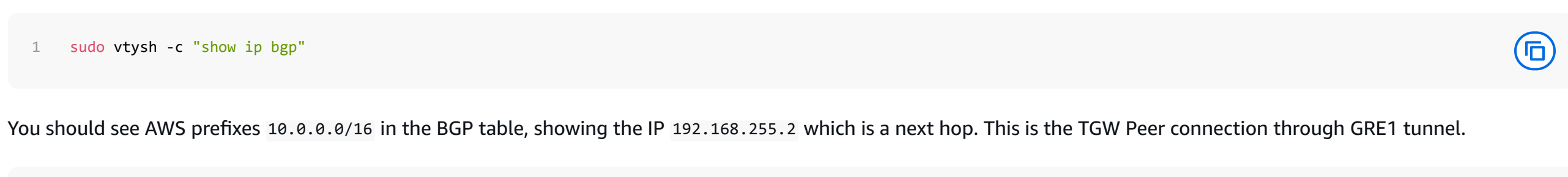
Now that the routing tables are correctly configured, verify that the **Third-Party Appliance** is receiving **dynamic routes** from **AWS** via **BGP**.

Connect to the Appliance via Session Manager

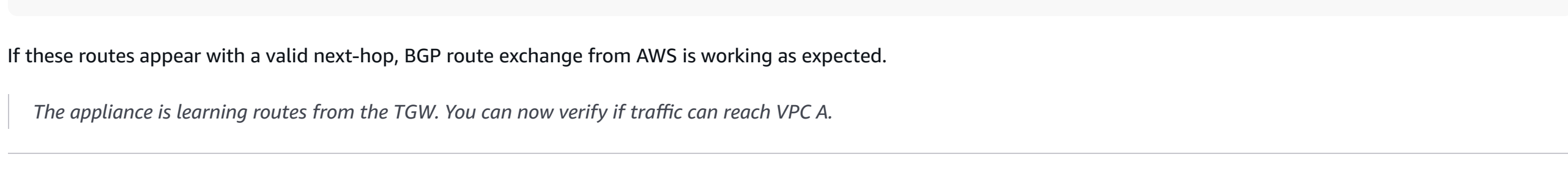
1. Go to the [EC2 Console](#).
2. Locate the instance named **Third-Party Appliance**.
3. Select the instance, click **Connect**, then go to the **Session Manager** tab.
4. Click **Start session**.

Inspect BGP Routes

Once connected, run the following command to return all routes learned from BGP routing in the Third-party Appliance route table:



You should see **AWS prefixes 10.0.0.0/16** in the BGP table, showing the IP **192.168.255.2** which is a next hop. This is the TGW Peer connection through GRE1 tunnel.



If these routes appear with a valid next-hop, BGP route exchange from AWS is working as expected.

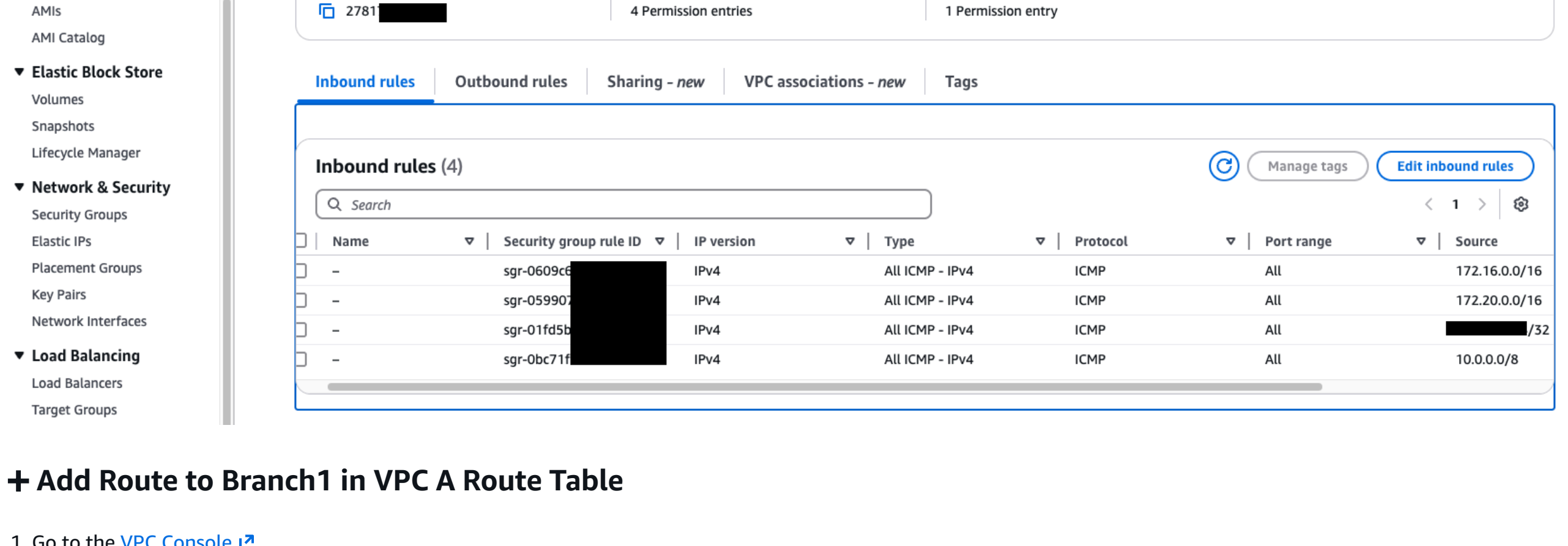
The appliance is learning routes from the TGW. You can now verify if traffic can reach VPC A.

Step 4: Test Reachability between Branch1 Router

This test verifies that **VPC A** is reachable from the **Third-Party network** via the Transit Gateway Connect Attachment and BGP-learned routes. We will ping the instance **VPC A Private AZ1 Server** from the **Branch1 Router**.

Ensure VPC A Instance Allows ICMP

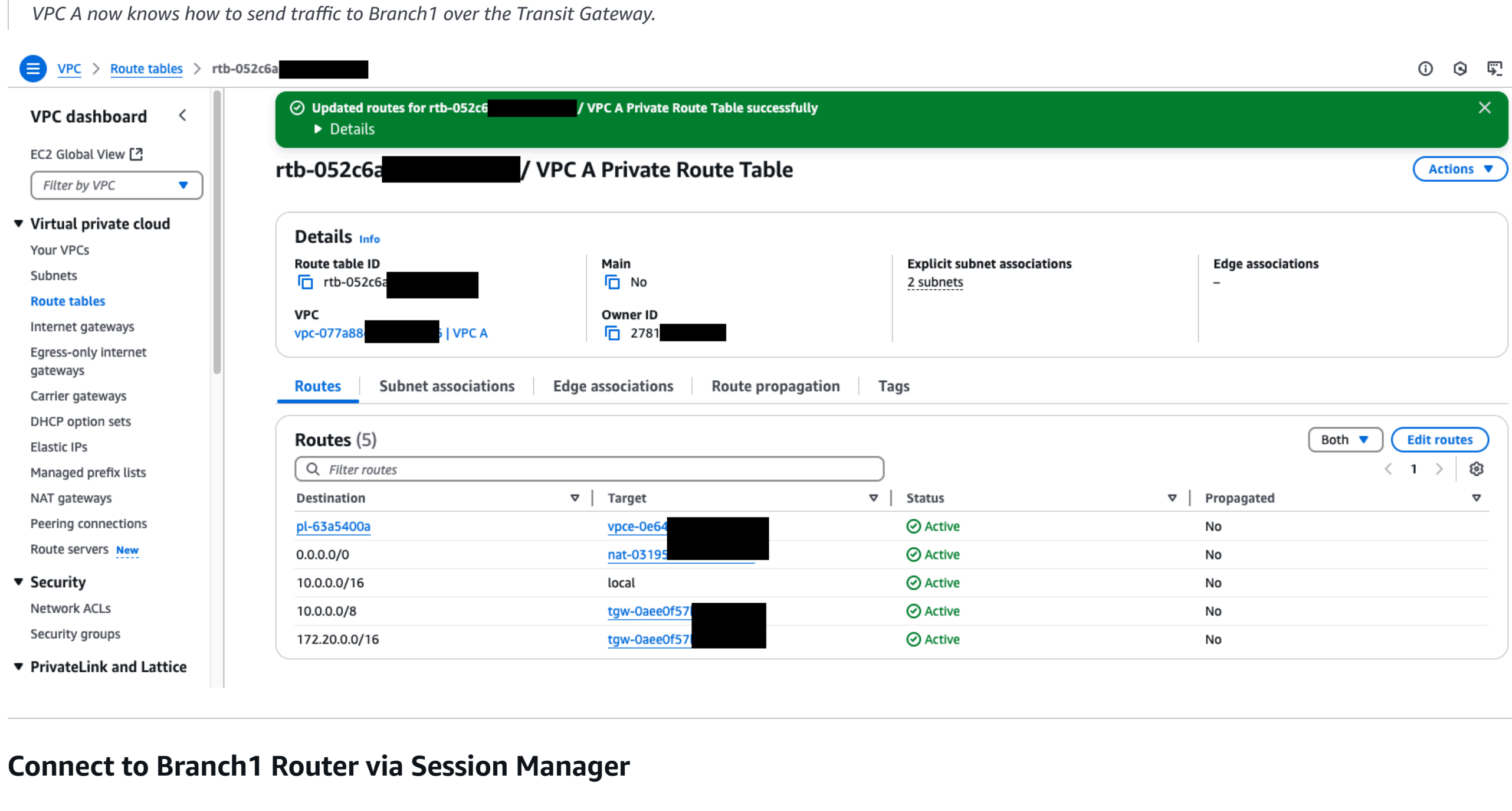
1. Open the [EC2 Console](#).
2. Find the instance named **VPC A Private AZ1 Server**.
3. Note its **private IP address**. The IP was statically assigned during the instance creation to **10.0.1.180**.
4. Select the instance, then go to the **Security** tab.
5. Click the linked **Security Group**.
6. Under **Inbound rules**, click **Edit inbound rules**.
7. Add a new rule:
 - **Type**: All ICMP - IPv4
 - **Source**: 172.20.0.0/16 (Branch1 VPC CIDR)
8. Click **Save rules**.



+ Add Route to Branch1 in VPC A Route Table

1. Go to the [VPC Console](#).
2. Locate the route table named **VPC A Private Route Table**.
3. Select the route table and go to the **Routes** tab.
4. Click **Edit routes**.
5. Add a new route:
 - **Destination**: 172.20.0.0/16
 - **Target**: Transit Gateway -> select the TGW attachment **VPC A Attachment** from the dropdown list.
6. Click **Save routes**.

VPC A now knows how to send traffic to Branch1 over the Transit Gateway.



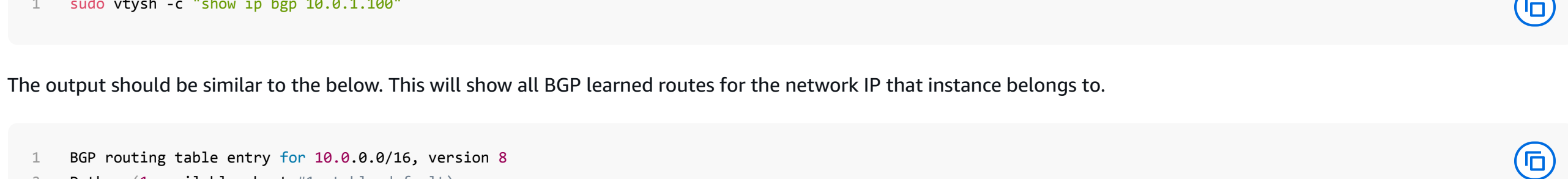
Connect to Branch1 Router via Session Manager

1. Return to the [EC2 Console](#).
2. Find the instance named **Branch1 Router**.
3. Click it and select **Connect**.
4. Choose **Session Manager** and click **Start session**.

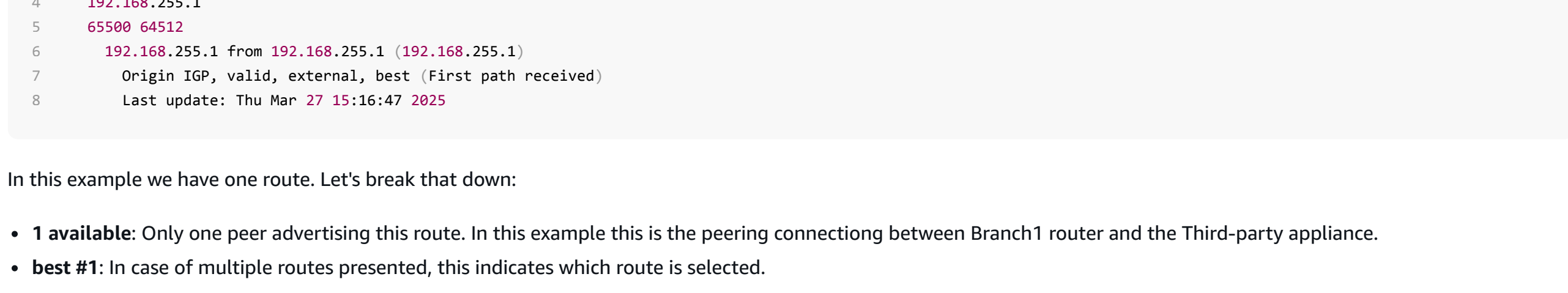
Inspect the Learned Routes to VPC A

Before running the ping test, verify that the **Branch1 Router** has learned the route to the **VPC A Private AZ1 Server** over **BGP**.

In the terminal session of **Branch1 Router**, run the following command to return any routes learned from BGP peers for the private IP of **VPC A Private AZ1 Server**:



The output should be similar to the below. This will show all BGP learned routes for the network IP that instance belongs to.

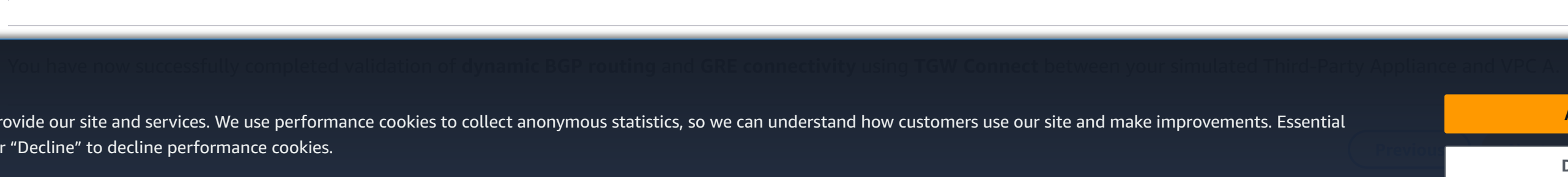


In this example we have one route. Let's break that down:

- **1 available**: Only one peer advertising this route. In this example this is the peering connection between Branch1 router and the Third-party appliance.
- **best #1**: In case of multiple routes presented, this indicates which route is selected.
- **192.168.255.1**: The peer that advertised this route to Branch1.
- **64512**: is the ASN of the Transit Gateway (TGW).
- **65500**: is the ASN of the Third-Party Appliance.

Run Ping Test

In the terminal session of **Branch1 Router**, run:



Since we allowed the network **172.20.0.0/16** in the Security Group for the instance **VPC A Private Route Table**, we are using the source IP **172.20.0.5** of the directly connected interface for the ICMP testing.

You should receive successful replies — confirming end-to-end routing from the Third-Party network to VPC A is working dynamically via the TGW Connect Attachment.

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